UNIVERSITY OF BRITISH COLUMBIA • SCHOOL OF KINESIOLOGY

KIN GRADUATE RESEARCH DAY

Program & Abstract Booklet

May 7, 2026

Gateway Room 1004 & B1102

University of British Columbia

Kinesiology Graduate Student Society (KGSS)

Land Acknowledgement

The Kinesiology Graduate Student Society respectfully acknowledges that the Kinesiology Graduate Research Day is hosted on the traditional, ancestral, and unceded territory of the xʷməθkʷəy̓əm (Musqueam) People. We are grateful to the Musqueam people for their enduring stewardship of these lands, and we recognize the responsibility we share in supporting and respecting Indigenous communities – past, present, and future.

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KIN Graduate Research Day

PROGRAM

May 7, 2026 • Gateway Room 1004

MORNING

8:00 – 8:45 AM	**Registration & Breakfast**
8:45 – 8:55 AM	**Welcome & Opening Remarks**
8:55 – 9:00 AM	**KTEP Presentation**

Session 1 — 3-Minute Thesis 9:00 – 9:35 AM

Time	Presenter | Title
9:00 – 9:06	**Sayna Soleymanian**, Kyle J. Missen, Allan L. Adkin *The effects of motivational self-talk on postural threat-induced changes in emotions, attention focus, and static and dynamic balance control*
9:06 – 9:12	**Quaila Trang** *Sensory contribution to the preparation and control of movement*
9:12 – 9:18	**Katy Stewart** *A Longitudinal Narrative Analysis of Resilience in Competitive Athletes*
9:18 – 9:24	**Matt Galassi** *Development of a Novel Perturbation and Use of Indwelling EMG for Motor Threshold Detection in Neck Muscle Architecture*
9:24 – 9:30	**Molly Strain**, Yao Sun, Tania Lam *The effects of different tibial neurostimulation patterns on pelvic floor muscles*
9:30 – 9:35	**Megan Tomyn** *Indigenous Student-Athlete Representation in Canadian Universities: Addressing Data Gaps and Promoting Equity*

9:35 – 9:45 AM	*Break (10 min)*

Session 2 — 10-Minute Oral 9:45 – 11:00 AM

Time	Presenter | Title
9:45 – 10:00	**Annika Szarka**, Nick Butler, Elsa Sun, Romeo Chua *Does hand-to-target information contribute to implicit adaptation of target-directed reaching?*
10:00 – 10:15	**Samantha Graham** *Imaging disability: Visual representations of disability at the School of Kinesiology in the UBC Archives*

10:15 – 10:30	**Yasmeen Jalil**, Makenna Cameron, Daniel Gamu *Stable expression of histone demethylase JMJD3 in mouse skeletal muscle with treadmill training*
10:30 – 10:45	**Carmen Golnaraghi** *UBC TogetherActive: Supporting Connections Through Accessible Recreation*
10:45 – 11:00	**Xiangwei Zhang**, Ivan Wong, Mark Carpenter *Transfer of Postural and Emotional Adaptation Following Graded Virtual Height Exposure to Real Heights*

11:00 – 11:15	**Break (15 min)**
11:15 AM – 12:30 PM	**INTERDISCIPLINARY PANEL** Dr. Jesse Charlton • Dr. Jasmin Ma • Dr. Cailie McGuire • Dr. Janice Forsyth • Dr. Daniel Gamu

LUNCH & POSTER VIEWING • 12:30 – 1:30 PM • Gateway B1102

- **Gabrielle Yarema**, Cameron Mitchell, Lucas Wiens, Haley Dennis, Matthew Fliss
  Limitations to work capacity during low- and high-load multi-joint resistance exercise
- **Nesha Colgate**, Jamie Bell, Jem Arnold
  Validation of Tymewear-Reported Minute Ventilation and Breathing Rate During Laboratory-Based Cycling Exercise
- **Carrie Peters**, Christopher J.H. Gray, Romeo Chua, Sarah N. Kraeutner, Nicola J. Hodges
  The contribution of observed kinematics and outcomes in observation-induced contagions in pointing movements

AFTERNOON

Session 3 — 10-Minute Oral 1:30 – 3:00 PM

Time	Presenter | Title
1:30 – 1:45	**Imen Zouaoui**, Makenna S. Cameron, Daniel Gamu *Role of CBP/p300 in BAT adaptive thermogenesis and systemic energy metabolism*
1:45 – 2:00	**Liam Foulger**, Jesse Charlton, Jean-Sébastien Blouin *Our head is most stable at our preferred walking and running speeds*
2:00 – 2:15	**Samantha Mew** *A Lifetime of Exposure: Insights into Women's Realities of Endocrine Disrupting Chemicals (EDCs) in Vancouver, Canada*
2:15 – 2:30	**Jason Mansour Anasori** *Targeting spinal convergence in pelvic floor muscles through timing-dependent paired stimulation*

2:30 – 3:00	**Ally Schweitzer**, Max Abercrombie, Justin Losciale, Cameron Mitchell *Muscle collagen accumulation is not a universal feature of human aging: a systematic review*
2:45 – 3:00	**Jack Darley** *Where am I? The influence of prediction on localizing the unseen hand*

3:00 – 3:15 PM	**Break (15 min)**

Session 4 — 3-Minute Thesis 3:15 – 3:51 PM

Time	Presenter | Title
3:15 – 3:21	**Winston Brandt** *Exploring Community Gym Attendance and Aerobic Fitness in Spinal Cord Injury*
3:21 – 3:27	**Nick Butler** *The mechanisms driving tactile facilitation with the expectation of object reception*
3:27 – 3:33	**Daniel Everard** *The influence of sensorimotor uncertainty in movement decision-making*
3:33 – 3:39	**Grace French** *Clear Roles, Strong Teams: A Social Network Analysis of Role Clarity in Sport*
3:39 – 3:45	**Emmarie Racine Hallin**, Josh Donald *Age-related differences in vestibular perception*
3:45 – 3:51	**Nada Shweiki**, Liam Foulger *Optimizing Noisy Vestibular Stimulation for Maximizing Sway and Increasing Postural Oscillations*

3:51 – 4:01 PM	*Break (10 min)*

Session 5 — 10-Minute Oral 4:01 – 5:00 PM

Time	Presenter | Title
4:01 – 4:16	**Xueqing Zhou**, Alison Williams, Jiho Song, Jeff Nickel, Tania Lam, Jean-Sébastien Blouin *What can we learn about the pelvic floor muscles on a (pseudo) roller coaster ride?*
4:16 – 4:31	**Xiya (Christina) Cheng**, Xiangwei Zhang *Height-induced postural threat and its influence on upper limb fine motor control*
4:31 – 4:46	**Ellie Saplywy** *Exploring motivations for attendance and fanship in Canadian women's university sport*
4:46 – 5:01	**Amin MohammadiNasrabadi**, Pedro Amui, Izzy Zhang, Calvin Kuo, Jean-Sébastien Blouin

	Computational models to explain the role of efferent copies in standing balance with delays
~5:00 PM	**Closing Remarks**

KIN Graduate Research Day

ABSTRACT BOOKLET

May 7, 2026 • Gateway Room 1004

Session 1 - 3-Minute Thesis • 9:00 – 9:35 AM

1. *9:00 – 9:06*

The effects of motivational self-talk on postural threat-induced changes in emotions, attention focus, and static and dynamic balance control

Sayna Soleymanian, Kyle J. Missen, Allan L. Adkin

Supervisor: Dr. Mark G. Carpenter

The effects of fear of falling on balance have been investigated through manipulating surface height (i.e., height-induced postural threat) which has led to maladaptive emotional and postural responses. Motivational self-talk (ST) has successfully improved athletes' performance and reduced anxiety in sports. Given its inconsistent results in balance research, it is unknown whether motivational ST will mitigate the maladaptive responses observed with height-induced postural threat.

Thirty-eight healthy young adults were randomized to two groups (motivational and neutral ST). Participants stood on a forceplate at the edge of a hydraulic platform, and completed quiet stance and rise to toes tasks at low (0.8 m) and high heights of 3.2 m (quiet stance) and 1.6 m (rise to toes). During each trial, centre of pressure was calculated from the forceplate, along with bilateral tibialis anterior and soleus activity, and electrodermal activity (a measure of physiological arousal). Participants also completed state questionnaires to assess their emotional states.

There were mainly robust height-induced responses observed for both tasks with regards to emotional, autonomic, and balance control measures. Engaging in motivational ST during quiet stance at high compared to low heights led to a significantly smaller increase in postural sway frequency. Speaking (motivational or neutral ST) during quiet stance led to smaller increases in directing attention toward self-regulatory strategies at height. There were no significant interactions between height and ST for rise to toes.

This thesis study was the first study that combined engaging in motivational ST with height-induced postural threat for quiet stance and rise to toes tasks. Engaging in motivational ST was largely unsuccessful at attenuating the maladaptive responses observed with height-induced postural threat; however, motivational ST is still a powerful tool which may be successful at improving balance control for older adults, various clinical populations, or in milder forms of postural threat.

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2. *9:06 – 9:12*

Sensory contribution to the preparation and control of movement

Quaila Trang (she/her)

Supervisor: Dr. Romeo Chua

Precise control of voluntary goal-directed movements depends on incoming positional and velocity-based sensory information, termed kinesthetic feedback. A challenge for the central nervous system (CNS) is the inherent delay in this sensory transmission. Despite this delay, we can routinely execute rapid, accurate movements. One theory posits that the CNS monitors state of the limb by preferentially weighting the intended motor output until incoming sensory feedback is available. Research has shown that in the control of target-directed movements, the CNS uses afferent information regarding initial limb position during motor planning (Gordon et al., 1995; Sainburg et al. 2002), while preferentially weighting afferent signals from lengthening muscles to estimate ongoing limb position and velocity during execution (Eschelmuller et al. 2025). A

question of interest is how feedback regarding initial limb position influences the planned extent of a movement when incoming movement-related feedback would be available during execution of the movement.

This proposed study is designed to investigate how performance of a target-directed movement is influenced by the potential utility of sensory feedback during both the planning and execution of the movement. Participants will be required to make unseen elbow extension movements to a visual target. Muscle vibration applied to the biceps, triceps or both prior to the start of movement, will be used to influence kinesthetic perception of initial limb position. Movement duration (fast-ballistic: 150-250ms; slow-targeting: 800-900ms) will be varied to influence the time available to process and use movement-related feedback.

The analysis of movement performance under varied sensory conditions and task demands is a means to investigate how our sensorimotor systems utilize and preferentially weight the contributions of efferent motor output and afferent sensory signals. Specifically, how the weighting of these signals change during the preparation and control of voluntary goal-directed movements to adapt to changing sensorimotor conditions.

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3. *9:12 – 9:18*

A Longitudinal Narrative Analysis of Resilience in Competitive Athletes

Katy Stewart (she/her)

Supervisor: Dr. Desmond McEwan

The best athletes in the world fail all the time. They lose points, miss shots, and fall short of expectations. Roger Federer, one of the tennis greats, highlights that the mark of a champion is someone who knows they will lose repeatedly and has learned to deal with it. This idea is captured by the concept of resilience: a dynamic social process which individuals and groups engage in when faced with adversity. Despite more than 20 years of sport research on resilience there still exists a considerable lack of conceptual clarity. There have been repeated calls for longitudinal work, however, there has yet to be an exploration of resilience in sport over time.

Thus, the objective of my proposed research is to explore athletes' experiences of resilience over time through the stories they tell. This research will be conducted using narrative inquiry, which involves studying stories as they evolve over time. Narratives are particularly useful for capturing both personal and social meanings, as well as accounting for temporality. I will approach this work from a social constructionist worldview, recognizing that experiences of adversity, and thus resilience, are unique to the individual and influenced by social contexts. I plan to explore competitive athletes' stories of resilience over 8 weeks through a combination of semi structured interviews and weekly participant diaries. Using narrative thematic analysis, I aim to identify themes among the stories that athletes share to better understand what resilience looks like in the context of competitive sport.

This research will make an important theoretical contribution as the first longitudinal study of resilience in sport and lay the foundation for future work. There are also potential applied implications: the participants themselves may benefit from the experience of completing diaries and the stories told may provide applied sports psychology practitioners with insights to inform their practice. In summary, this research takes a novel approach to exploring resilience in sport and stands to contribute both theoretically and practically.

4. *9:18 – 9:24*

Indwelling and High Density Surface EMG for Neck Muscle Motor Threshold Detection in Response to Novel Angular Perturbations

Matt Galassi

Supervisor: Dr. Jean-Sébastien Blouin

Introduction:
Despite the importance of neck muscles for movement control and head neck injury, major breakthroughs are limited due to their multi-layered anatomy and the impossibility to voluntary activate individual neck muscles. This project investigates how individual neck muscles generate compensatory activation in response to angular torso perturbations across varying velocities and directions, focusing on identifying motor thresholds.

Methods:
Healthy participants will be seated on a KUKA robotic arm programmed to deliver angular perturbations centered at the C5-C6 spinal level, allowing the head to remain free in space while instructed to keep their head upright. Dynamic testing will use sinusoidal velocity profiles in pitch and roll, and a 45 degree combination of both, performed under two conditions: eyes open and eyes closed. Neck muscle activity will be recorded using indwelling fine-wire EMG electrodes in the sternocleidomastoid, and splenius capitis, and a 65 channel high density surface EMG to compare the difference in threshold detection. Inertial measurement units (IMUs) placed on the chest, forehead, and the KUKAs flange, will capture body and head kinematics.

Results:
We predict that, above a velocity-dependent motor threshold, muscle activation will generate antiphasic head responses to torso angular velocity stabilizing the head upright, with this relationship breaking down below threshold, becoming in-phase. The motor threshold and its velocity-dependency will be quantified using signal detection theory (d') on enveloped EMG relative to angular velocity. We hypothesize that the antiphasic compensatory pattern will be detected at lower velocities with eyes open, and higher with eyes closed due to reduced contribution to sensory integration.

This study highlights the interplay between vestibular, velocity proprioception, and visual inputs in head-neck stabilization, advancing understanding of the VCR and CCR reflexes. This will provide insight into the input-output relationships of neck muscle control, informing future test designs.

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5. *9:24 – 9:30*

The effects of different tibial neurostimulation patterns on pelvic floor muscles

Molly Strain, Yao Sun, Tania Lam

Supervisor: Dr. Tania Lam

Overactive bladder is a type of lower urinary tract dysfunction that causes sudden and intense urges to urinate that are often difficult to control and lead to incontinence. Medications and catheterization are the main management strategies for this condition in neurogenic populations, however they come with adverse side effects. Posterior tibial nerve stimulation (PTNS), which applies mild electrical stimulation near the ankle, is a promising alternative treatment with no obvious side effects. Previous clinical studies on PTNS predominantly used protocols where stimulation is applied continuously. However, emerging evidence suggests that phasic inputs may better modulate spinal excitability.

This study compared the effects of different stimulation patterns applied to both legs in healthy participants. Each participant completed three PTNS sessions (continuous, intermittent, and patterned) conducted over separate visits and PFM reflex responses were measured immediately before and after each session. The continuous condition consisted of uninterrupted stimulation, while the intermittent and patterned conditions used a 600 ms ON: 400 ms OFF duty cycle. In the patterned condition, stimulation alternated between the left and right sides with 100 ms of bilateral overlap. We hypothesized that patterned stimulation would produce the greatest facilitation due to enhanced sensory feedback similar to those produced during walking.

Data from seven participants suggested that PTNS was associated with an unexpected decrease in the tibial-PFM reflex response, with no differences between conditions. Future research is warranted to further understand how PTNS affects bladder control and support the development of more effective, non-invasive treatments for people with neurological conditions.

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6. *9:30 – 9:35*

Indigenous Student-Athlete Representation in Canadian Universities: Addressing Data Gaps and Promoting Equity

Megan Tomyn (She/Her/Hers)

Supervisor: Dr. Janice Forsyth

Indigenous representation in Canadian university sport remains poorly understood, despite growing attention to Indigenous enrollment in post-secondary education. At the University of Saskatchewan (USask), only 3% of student-athletes self-identify as Indigenous, compared to 12.7% of the overall student body (University of Saskatchewan, n.d.; Tomyn & Schaefer, 2025). The absence of disaggregated data reflects systemic invisibility, limiting institutions' ability to assess equity, identify barriers, and track progress toward reconciliation in sport (Forsyth et al., 2023; Smith, 2012). Colonization disrupted Indigenous physical activity systems, replacing them with Eurocentric models that continue to shape post-secondary sport (Forsyth, 2020; Robidoux & Mason, 2017). Although the Truth and Reconciliation Commission of Canada (2015) calls for inclusive sport policies, Indigenous participation remains under-documented and under-supported.

This research addresses the data gap in two phases. Phase one employs a cross-sectional, quantitative analysis of Indigenous student-athlete participation across U15 Canadian universities, disaggregated by First Nations, Métis, and Inuit identities. Phase two uses semi-structured interviews with student-athletes, teachers, coaches, and community leaders to examine perceived barriers, pathways, and culturally relevant strategies for fostering inclusion. Both phases are grounded in Indigenous research methodologies emphasizing relational accountability, narrative sovereignty, and ethical engagement (Kovach, 2009; Wilson, 2008).

By integrating quantitative and qualitative approaches, this project will generate the first national dataset on Indigenous student-athlete representation while centering Indigenous voices in sport. Findings will inform institutional policies, athletic programming, and national strategies to advance equity, inclusion, and reconciliation in Canadian university athletics.

Session 2 - 10-Minute Oral • 9:45 – 11:00 AM

7. *9:45 – 10:00*

Does hand-to-target information contribute to implicit adaptation of target-directed reaching?

Annika Szarka (she/her), Nick Butler, Elsa Sun, Romeo Chua

Supervisor: Dr. Romeo Chua

Implicit sensorimotor adaptation underlies our ability to subconsciously adjust our movements to varying sensorimotor conditions. During target-directed reaching movements, our perception of where our hand is in space is based on the brain's multimodal representation of hand position. Computational models have suggested that the difference between this perceived hand position and the target is a critical error signal in driving adaptation. Despite the significant contribution of actual hand position information towards perceived hand position, many studies use "shooting" movements that reach beyond the target such that the actual hand position is no longer near the target at the end of movement. Currently, no study has compared the difference in implicit adaptation between shooting and pointing movements where the hand is directed at and to the target.

We recruited 30 participants who performed reaching movements with their unseen hand towards targets presented on a display. Visual feedback of hand position was provided by a cursor on the display. To induce adaptation, cursor feedback was rotated (clamped) relative to the target position, irrespective of reach direction. Participants performed two sessions one-week apart: one with shooting and one with pointing movements. Based on current computational models of adaptation, we hypothesized that pointing movements would generate a more precise proprioceptive cue relative to the visual target and draw the perceived hand position closer to the target. This would lead to a smaller error signal and subsequently less adaptation.

Contrary to our hypothesis, there was no difference in adaptation between the pointing and shooting conditions. We did observe that adaptation decreased in the second session regardless of what condition was being performed. These results prompt further examination of the type of information that underlies sensorimotor adaptation in goal-directed movements and of changes in adaptive processes with re-exposure to previously novel sensorimotor conditions.

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8. *10:00 – 10:15*

Imaging disability: Visual representations of disability at the School of Kinesiology in the UBC Archives

Samantha Graham

Supervisor: Dr. Andrea Bundon

Images are never innocent. Visual representations of disability are never completely unbiased, and can only be partially reflective of the rich, storied lives led by disabled people. In the same way, archival images are never unmediated representations of the past, but are always incomplete, unfinished, and open to (re)interpretation and activation in the present. Researchers have noted that disabled people are often missing from the archives, both past and present. Similarly, studies in kinesiology have found that disabled people are underrepresented in the field. To address this present absence, recent scholarship has called for critical and creative engagement with equity, diversity, inclusion, and accessibility in both kinesiology and the archives.

My proposed research aims to contribute to the literature by exploring how disability is visually represented at the School of Kinesiology in the UBC Archives between 1946 and 2011. Drawing from critical disability studies and archival research literature, the research will use critical discourse analysis to understand these visual texts and the social construction of disability. A critical visual methodology will look at the image, the spectator, and the complex system of social and cultural relations that both are embedded in.

Current social and cultural conditions are a product of our collective past, as (en)countered and (re)constructed in the archive; as such, the archived past determines the conditions of the present, and therefore the possibility of the future. How could disability be imag(in)ed otherwise in Kinesiology at UBC?

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9. *10:15 – 10:30*

Stable expression of histone demethylase JMJD3 in mouse skeletal muscle with treadmill training

Yasmeen Jalil (she/her), Makenna Cameron, Daniel Gamu

Supervisor: Daniel Gamu

Skeletal muscle is highly plastic and capable of vast contractile and metabolic remodelling by exercise, which is supported by activation and repression of various gene programs. Such remodelling is coordinated, in part, by post-translational modifications to histones, and thereby chromatin structure. One key residue is lysine 27 on histone 3 (H3K27), which can differentially alter chromatin accessibility and transcription. Methylation of H3K27 (H3K27me1-3) causes chromatin compaction and transcriptional repression whereas demethylation facilitates transcriptional activation. Several methylated histone marks and their respective modifying enzymes can be changed by exercise training. As a key transcriptional switch, H3K27 may play a vital role in facilitating exercise-induced cellular adaptations. As such, we hypothesized that the expression of various H3K27 demethylases (e.g. UTX, UTY, JMJD3, JHDM1D) would be altered in skeletal muscle by endurance training.

To address this, male and female mice (C57BL/6J; n = 5-6/group) were progressively treadmill trained for 8 weeks and compared to sedentary littermate controls. Body composition was measured prior to and following the training, along with body mass throughout. Skeletal muscles of differing contractile/metabolic properties were collected 48-hours after the final exercise bout. Training improved body composition of female exercisers only. As expected, trained mice of each sex had increased expression of mitochondrial marker proteins in the red/white gastrocnemius and red quadriceps. However, JMJD3 expression was unchanged in either sex.

Moving forward, we will analyze remaining tissues for UTX, UTY, JMJD3, and JHDM1D expression. Additionally, we aim to explore how the sub-cellular localization of these demethylases might change with exercise training. Our work will help clarify how chromatin dynamics participate in the regulation of skeletal muscle exercise adaptations.

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10. *10:30 – 10:45*

UBC TogetherActive: Supporting Connections Through Accessible Recreation

Carmen Golnaraghi (She/her)

Supervisor: Dr. Erica Bennett

In 2025, approximately 20.7 percent of students at the University of British Columbia (UBC) reported having a disability. Students commonly reported visual impairments, functional limitations

of the optical system that restrict eyesight despite correction. In 2022, UBC created its "Calls to Action to Enable Meaningful Implementation of the Accessible British Columbia Act" to ensure that students with visual impairments are academically supported on campus. While this connects students with visual impairments to free tutors, notetakers, and safe study spaces, it fails to provide sufficient resources to promote their sport participation.

UBC Recreation and Athletics offers over 50 physical activity programs and 11 recreational sport leagues intended to promote student well-being and community. However, findings from the 2025 Student Diversity Census indicate fear and disengagement among students with visual impairments. This gap must be addressed, as limited participation in campus events is a contributor to psychological distress.

The purpose of this study is to launch "UBC TogetherActive," a pilot program advancing inclusive physical activity for students with visual impairments. In partnership with BC Blind Sports, students are paired with trained volunteers for weekly sessions to co-navigate campus fitness spaces. This study examines: 1) How participants interpret belonging and safety within campus recreation after the program, and 2) What recommendations they have for improving accessibility within UBC recreation. Adopting a relativist ontological and subjectivist epistemological stance, I will conduct semi-structured interviews with 5 to 8 students with visual impairments and 5 to 8 volunteers post-participation. The outcome of this research is to promote the long-term independent participation of students with visual impairments in recreational fitness.

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11. *10:45 – 11:00*

Transfer of Postural and Emotional Adaptation Following Graded Virtual Height Exposure to Real Heights

Xiangwei Zhang, Ivan Wong, Mark Carpenter

Supervisor: Mark Carpenter

Virtual Reality (VR) has shown utility in examining postural adjustments related to fear, anxiety, and balance confidence at elevated heights. However, the degree to which emotional and postural adaptations acquired in VR can transfer to real-world contexts remains unclear. This study examined emotional and standing balance adaptations following repeated exposure to graded virtual heights across two visits, and the subsequent transfer of these adaptations to real heights.

Thirty-six healthy young adults were randomly assigned to a VR or Control group. VR participants completed three visits separated by 24-72 hours, while Control participants completed two visits separated by 24-48 hours. In the first two visits, VR participants performed repeated quiet standing trials under graded virtual height conditions (0-50 m). Control participants completed repeated standing trials at a fixed low height (0.8 m) in a real environment. In the final visit, both groups performed quiet standing on a real elevated platform at low (0.8 m) and high (3.2 m) heights.

Results showed that the VR group exhibited threat-related emotional responses that adapted toward baseline levels across repeated virtual height exposures, whereas no such changes were observed in the Control group. During real-height exposure, both groups demonstrated increased fear, anxiety, and arousal, reduced balance confidence, and altered attentional focus at high compared to low height. However, group effects and interactions indicated that the VR group exhibited attenuated emotional responses and distinct postural control patterns relative to the Control group. These findings suggest that VR can effectively elicit and train emotional responses and certain threat-related postural strategies, particularly in graded, extreme elevated virtual height scenarios.

Poster Session - 12:30 – 1:30 PM • Gateway B1102

27. *Poster*

Limitations to work capacity during low- and high-load multi-joint resistance exercise

Gabrielle Yarema (she/her), Cameron Mitchell, Lucas Wiens, Haley Dennis, Matthew Fliss

Supervisor: Dr. Cameron Mitchell

There are important physiological differences between aerobic exercise (ex: running) and resistance exercise (ex: weightlifting). During aerobic exercise, the amount of work that can be performed (i.e. work capacity) is dependent upon how quickly active muscle can generate usable energy for contraction using the oxygen delivered from the blood. In resistance exercise, work capacity is dependent on the external load. Central (nervous system) and peripheral (local muscle) fatigue both contribute to determining work capacity, but their relative contributions as a factor of external load remain unclear. This study investigated the effect of external load, central fatigue, and peripheral fatigue on work capacity during multi-joint exercise.

Sixteen participants performed back squats to failure at 30%, 50%, 65%, and 80% 1RM. Pre- and post-exercise assessments of blood lactate, maximal voluntary contraction (MVC), voluntary activation (VA), and resting twitch (RT) quantified fatigue. Data were analyzed via mixed-model repeated measures with LSD post hoc comparisons.

Work capacity showed a significant inverse relationship with load ($p < 0.001$), peaking at 30% and lowest at 80%. Blood lactate changes were significantly lower at 80% ($p < 0.05$) compared to all other loads. MVC and resting twitch decreases were greatest at 30% and significantly higher than at 80% ($p < 0.01$). While VA significantly declined post-exercise ($p < 0.001$), no significant load or load × timepoint interactions were observed. Lower relative loads induce greater physiological disruption and peripheral fatigue due to higher total work volume. However, VA (central fatigue) appears to reach a similar threshold regardless of load.

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28. *Poster*

Validation of Tymewear-Reported Minute Ventilation and Breathing Rate During Laboratory-Based Cycling Exercise

Nesha Colgate, Jamie Bell, Jem Arnold

Supervisor: Dr. Michael Koehle

Minute ventilation (VE) and respiratory rate (RR) may be useful for intensity control during cycling training. However, current gold-standard devices are not practical for field use. Tymewear has developed a wearable chest-strap device, called Tymewear VitalPro (TW), intended to estimate VE and RR using plethysmography, making it a practical wearable device for athletes in the field. This study aimed to evaluate the operational validity and agreement of RR and VE measured by TW compared with the gold-standard metabolic cart, TrueOne 2400 (Parvo). We hypothesized that RR and VE measured by TW will demonstrate a strong correlation between devices. However, as VE reported by TW is in arbitrary units, strong agreement between devices would only be observed in RR.

16 participants (9M:7F) completed 5×5-min cycling interval workout at a self-selected intensity corresponding to the first ventilatory threshold (VT1). Metabolic data was collected using Parvo, while TW concurrently recorded RR and VE. Validity and agreement of RR and VE were analyzed using repeated measures correlation and Bland-Altman limits of agreement (LOA).

For RR, data analysis supported the study hypothesis, showing strong repeated measures correlation (r [95% CI]: 0.9 [0.9, 0.9], $p < 0.001$), and good agreement (bias -0.3 brpm, upper LOA=8.3 brpm, lower LOA=-8.9 brpm) between devices. For VE, there was only moderate repeated measures correlation ($r=0.470$, $p < 0.001$) and poor agreement (bias 41 L/min). Findings suggest that TW is valid for measuring RR during laboratory-based cycle ergometry and demonstrates good agreement with the gold standard, but VE agreement remains poor.

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29. *Poster*

The contribution of observed kinematics and outcomes in observation-induced contagions in pointing movements

Carrie Peters (she/her), Christopher J.H. Gray, Romeo Chua, Sarah N. Kraeutner, Nicola J. Hodges

Supervisor: Dr. Nicola Hodges

Watching the actions of others can cause unintentional performance biases. Seeing correct, yet different, actions can bias performance in the direction of the watched action. Compensatory contagions have also been identified in response to seeing errors. Differences between predicted and observed actions are thought to elicit a prediction error, updating the observer's internal model and causing later compensatory errors. However, evidence of compensation is limited and it is unclear what components of the observed action (i.e., kinematics or outcomes) underpin this effect.

To test for directional biases associated with observation and the conditions underpinning any biases, we conducted two experiments (N = 40 each). In a partial replication of a study where biases were shown, participants pointed to a vertically positioned centre target on a touchscreen computer and watched demonstrations of similar actions. Participants alternated blocks of pointing without visual feedback and watching videos of an actor aim with directional biases in three blocked conditions. Participants saw videos of a full action (kinematics), endpoints only (outcomes), or a control.

In Experiment 1, where the actor showed small biases, there was no evidence of imitative or compensatory biases across the groups or conditions. In Experiment 2, we increased the salience of the observed errors by showing larger actor biases. Imitative biases were present for the 'correct' group in both the kinematics and outcomes conditions, which did not differ, but not in the control condition. There were no biases shown in any condition by the 'error' group. The absence of compensation effects questions both the reliability of such an effect from watching, as well as the PE mechanism thought to underpin this effect.

Session 3 - 10-Minute Oral • 1:30 – 3:00 PM

12. *1:30 – 1:45*

Role of CBP/p300 in BAT adaptive thermogenesis and systemic energy metabolism

Imen Zouaoui (she/her), Makenna S. Cameron, Daniel Gamu

Supervisor: Daniel Gamu

Brown adipose tissue (BAT) is specialized for energy expenditure through expression of uncoupling protein 1 (UCP1), which dissipates the mitochondrial proton gradient to generate heat to regulate body temperature and energy balance. BAT structure and function are highly adaptive and are driven, in part, by epigenetic mechanisms. We previously showed that inducible BAT-specific double knockout (DKO) of the functionally homologous histone 3 lysine 27 acetyltransferases p300 and CBP results in marked reductions in BAT mass and thermogenic markers. *In vivo*, DKO mice developed rapid hypothermia within 2-4 hours during acute cold exposure compared to controls ($p<0.01$). Due to impaired thermogenesis in DKO mice, we tested their susceptibility to diet-induced obesity, hypothesizing increased adiposity and impaired glucose tolerance.

To induce BAT-specific knockout, mice carrying tamoxifen-inducible *Ucp1-CreERT2* were crossed with *Ep300/Crebbp* double-floxed mice yielding *Cre*-positive knockouts and floxed controls (n=5-10/group and sex). Mice were allocated to either a standard chow or a 'westernized' high fat diet (42% fat) for 16 weeks and singly-housed to monitor food intake. Glucose tolerance following a 5-hour fast and body composition analysis (EchoMRI) was conducted before and after the diet, and body mass was measured weekly. While our study is ongoing, we have found that loss of p300 and CBP results in mild glucose intolerance prior to diet initiation ($p <0.05$). Surprisingly, 16-weeks of a high-fat feeding did not reveal increased susceptibility to obesity and gluco-regulatory dysfunction in knockout mice. Further experiments are aimed at interrogating potential compensatory mechanisms explaining lack of an energy balance phenotype in these mice.

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13. *1:45 – 2:00*

Our head is most stable at our preferred walking and running speeds

Liam Foulger, Jesse Charlton, Jean-Sébastien Blouin

Supervisor: Dr. Jean-Sébastien Blouin

Accurate and reliable sensing of the world around us is fundamental to our ability to move and navigate. One way in which we maintain the quality of our sensory information as we move is by keeping our head stable, i.e., reducing the variability of our head movement. This is important because our head contains three important systems for sensing our self-motion: the visual, vestibular, and auditory systems. We previously found that head motion variability is lowest near preferred walking speeds, suggesting that sensory demands may play a role in shaping our movement patterns.

To test whether this extends to other locomotor behaviors, we examined whether preferred running speed also coincides with minimal head motion variability and how head motion variability changes as we transition between walking and running (~2 m/s). Five participants (2 F; 3 M) walked and ran outdoors across a wide range of speeds (<0.5 to >5 m/s). We quantified head movement variability as the stride-by-stride deviation from average head linear accelerations and angular velocities, and fit second-order polynomials to capture how it changed across speeds.

For linear acceleration, fits were strong for both walking ($R^2=0.97$, minima ~1.5 m/s) and running ($R^2=0.95$, minima ~2.9 m/s), with the two curves intersecting near 1.9 m/s. These are consistent

with the speeds that humans prefer to walk, run, and transition between walking and running, respectively. These results suggest humans select gait and transition speeds that keep our heads stable, like previously established preferences to move at speeds that minimize energy expenditure.

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14. *2:00 – 2:15*

A Lifetime of Exposure: Insights into Women's Realities of Endocrine Disrupting Chemicals (EDCs) in Vancouver, Canada

Samantha Mew (she/her)

Supervisor: Dr. Liv Yoon

Industrial activities create endocrine-disrupting chemicals (EDCs) that negatively affect human hormonal systems resulting in cancer, reproductive, and epigenetic diseases. Social determinants of health (SDOH) such as gender, race, and class shape EDC exposure risk, resulting in disproportionately higher harm for women facing multiple axes of marginalization. Current Canadian policies and proposed solutions frame EDCs as an issue of individual consumption and behaviour. With the rise in viewing EDC mitigation as an individual responsibility, the political, economic, and social relationships that enable EDC production are increasingly left unquestioned.

However, in response to the growing body of evidence showing disproportionate harm these policies may have led to for certain groups of women, there is a call for action from scholars to transition towards advocating for mitigation policies around EDC production that can protect all women. Using collage followed by semi-structured interviews and focus group sessions, this study aims to explore how women of various social positions in Vancouver, Canada experience EDCs, with a focus on the sociopolitical and economic structures that may have led to disproportionate risk in the first place for some.

The aim is to inform thinking about policy solutions that expand beyond consuming the 'right' products to include the structural issues that originally created the disproportionate EDC exposure and associated risks. In turn, the goal is to push for structural, regulatory changes in order to recognize and mitigate the uneven and unequal environmental risks that disproportionately harm women already facing multiple forms of marginalization.

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15. *2:15 – 2:30*

Targeting Spinal Convergence in Pelvic Floor Muscles Through Timing-Dependent Paired Stimulation

Jason Mansour Anasori

Supervisor: Dr. Tania Lam

Pelvic floor muscles (PFM) are essential for continence and are regulated by both cortical and spinal pathways, making them a relevant target for neuromodulation. Disruption of these pathways is common in neurological conditions, highlighting the need for approaches that can directly influence spinal motor circuits. Paired associative stimulation (PAS) uses precise timing between descending corticospinal activity and peripheral afferent input to induce activity-dependent plasticity and has been shown to strengthen motor pathways in limb muscles. However, its application to PFM remains largely unexplored, particularly at the level of spinal convergence.

The objective of this project is to determine whether PAS can modulate corticospinal excitability of the PFM by facilitating convergence of inputs at the sacral spinal cord, consistent with long-term potentiation-like effects. To test this, I control interstimulus intervals between transcranial

magnetic stimulation (TMS) applied over the motor cortex region controlling the PFM and pudendal nerve stimulation with millisecond precision using a custom Arduino-based triggering system. I record motor evoked potentials (MEPs) from the PFM using surface electromyography, and record spinal responses using a 15-channel lumbar electrode grid. To improve signal quality, I apply a spatial filtering pipeline that includes global common-average referencing, segment-specific mean subtraction, 2D Laplacian filtering, and generalized eigen decomposition.

This work establishes a methodological framework for applying PAS to pelvic floor motor pathways and provides initial evidence that temporally precise paired stimulation can modulate corticospinal excitability at the spinal level. These findings may inform the development of neuromodulation approaches targeting pelvic floor dysfunction in neurological populations, including individuals with spinal cord injury.

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16. *2:30 – 2:45*

Muscle collagen accumulation is not a universal feature of human aging: a systematic review

Ally Schweitzer (she/her), Max Abercrombie, Justin Losciale, Cameron Mitchell

Supervisor: Cameron Mitchell

The extracellular matrix is critical to skeletal muscle structure and function, with collagen its largest component. Fibrosis, excessive collagen accumulation, disrupts muscle function. Although animal studies consistently report age-related intramuscular collagen accumulation, human findings are inconsistent. This systematic review evaluated primary, peer-reviewed studies to assess if collagen accumulation is a universal feature of human aging.

The review was registered on PROSPERO. Following PRISMA guidelines, five databases were searched in January 2026 for studies comparing intramuscular collagen/extracellular matrix content in healthy young and older (>60 yr) adults. Eligible studies used histological or hydroxyproline techniques to quantify collagen/extracellular matrix content. Study screening, review, data extraction, and risk of bias were performed independently by two reviewers. Results were synthesized narratively.

Nine studies (including 122 young and 119 older adults) were included. Four reported no age-related differences, four showed age-related intramuscular collagen/extracellular matrix accumulation, and one found equivocal results when distinguishing perimysial from endomysial collagen. Considerable heterogeneity was observed in collagen quantification methods and control of mediators, including hypertension, diabetes, aerobic fitness, and physical activity. Studies with rigorous control generally found no age-related differences, whereas those with limited control generally reported age-related collagen accumulation. Collagen accumulation is not an inevitable feature of human chronological aging. Observed differences may instead reflect comorbidities or lifestyle factors associated with aging; thus, through these mediators, muscle collagen accumulation may be elevated in older populations. Future studies should control mediators and investigate mechanisms regulating collagen in skeletal muscle.

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17. *2:45 – 3:00*

Where am I? The influence of prediction on localizing the unseen hand

Jack Darley

Supervisor: Hyosub Kim

Whether sliding your arm through your jacket, or finding your ringing phone in a dark room, the sensorimotor system must precisely estimate the location of the unseen limb to move effectively. Proprioception, or the awareness of the state of the body, utilizes two sources of information: peripheral input from sensory receptors and central signals from motor commands. A copy of the motor command sent to spinal motor units is used to generate a motor prediction of movement outcomes and expected sensory consequences. Some studies contrasting active (voluntary) and passive (robot-guided) movement have shown that this prediction can be integrated with incoming sensory information to "enhance" proprioception. However, other studies have not, leaving overall evidence inconclusive.

Our proposed study will investigate these issues through the following two aims: (1) to understand how the presence of the motor prediction influences proprioceptive uncertainty, and (2) to elucidate the computational principles underlying localization of the unseen hand in 2D space. To this end, participants will either execute active reaching movements to a visual target, or be moved passively via a KINARM robot. After each reach, participants will indicate on a screen using a joystick-controlled cursor where they believe their unseen hand is.

We hypothesize that proprioceptive uncertainty will be reduced during active movement due to the presence of the motor prediction. Our pilot data (n=3) suggest that localization is best formalized as an optimal integration of prior expectations and incoming sensory information. Ultimately, these findings will quantify how active movement modulates proprioceptive precision, clarifying the computational principles underlying limb localization.

Session 4 - 3-Minute Thesis • 3:15 – 3:51 PM

18. *3:15 – 3:21*

Exploring Community Gym Attendance and Aerobic Fitness in Spinal Cord Injury

Winston Brandt (He/Him)
Supervisor: Tania Lam

Community-based fitness centres can help address critically low physical activity levels and elevated health risks in spinal cord injury (SCI) by providing opportunities to improve aerobic fitness. The six-minute arm test (6MAT) is a submaximal aerobic fitness measure; an individual exercises on an arm ergometer at a constant intensity, and heart rate (HR) is recorded in the final minute. Lower submaximal HR signifies better fitness, but large variability exists in SCI.

Between 2013-2016, 260 participants with SCI enrolled in an observational study at an accessible community fitness centre in Vancouver, BC. Physical assessments, including the 6MAT, were administered at intake and every 6 months. Facility attendance was recorded. Exploratory linear mixed-effects models were fit using univariate screening and bivariate building with a random intercept for participants.

Eighty participants had repeated 6MAT observations. The model included exercise intensity (watts/kg; $\beta=52.70$, $p<.001$), resting HR ($\beta=0.68$, $p<.001$), injury severity ($\beta=8.35$, $p=.009$), and visits per week ($\beta=-6.15$, $p=.021$); time, sex, and age were retained but non-significant. Injury severity and visits emerged as significant only after controlling for exercise intensity. Age was significant univariately but was absorbed by intensity, suggesting its effect operates through wattage selection. Each additional visit per week was associated with ~6 bpm lower HR. This project provides preliminary evidence for a relationship between self-directed community physical activity and improved aerobic fitness in SCI using a population health statistical approach.

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19. *3:21 – 3:27*

The mechanisms driving tactile facilitation with the expectation of object reception

Nicholas Butler
Supervisor: Dr. Romeo Chua

Sensitivity to tactile stimuli can change based on the sensorimotor context. Previous work has shown that sensitivity to tactile stimulation to the hand can be reduced immediately prior to and during a reach for an object. This modulation of sensitivity can further vary due to the expectation of contact with an object. Even in the absence of movement, sensitivity to tactile stimulation has been shown to change (increase) as a result of the expectation and visual feedback of an approaching object. For example, when receiving an object passed to us by another person, our sensitivity to tactile feedback is increased, presumably due to the importance of tactile information to indicate object contact or slip. The present study asks whether this modulation of tactile sensitivity is dependent only on the conscious expectation of object contact (i.e., reception) or whether it implicitly depends on the visual feedback being congruent with the expectation.

Participants are seated with their unseen hands resting on a table beneath a virtual display. A physical object below the display is brought into contact with the receiving hand, while virtual visual feedback of the object is shown concurrently on the virtual display. In one condition, the visual feedback of the approaching object is congruent (physical object and its visual feedback approach the right hand). In another condition, visual feedback of the object is incongruent (visual feedback shows object approaching right hand while participants explicitly expect that object is

actually brought to left hand). Cutaneous stimulation to the fingers is delivered at various intervals during the object movement to probe tactile sensitivity while awaiting reception.

The pattern of tactile sensitivity under these sensorimotor conditions will potentially reveal if the modulation is dependent only on conscious expectation of tactile input or implicitly requires veridical integration of visual and conscious information.

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20. *3:27 – 3:33*

The influence of sensorimotor uncertainty in movement decision-making

Daniel Everard (He/Him)

Supervisor: Mark Carpenter, Romeo Chua

Movement control can be viewed as a series of decisions made to maximize the utility of an outcome when faced with sensorimotor uncertainties; however, movement decisions can also be time-sensitive, leading to a trade-off between a person's speed and accuracy. While current literature demonstrates that humans are sensitive to the implicit variability of their motor system (i.e., risk-sensitive), limited research has looked at how humans optimally internalize their speed-accuracy trade-off throughout movement decision-making. Therefore, the proposed research seeks to explore how subjective risk-behaviour and movement aimpoint strategies are modified when movement speed and gain landscapes are manipulated during a reaching task. Participants will make reaching movements to a target line at varying speeds, with points linearly rewarded based on cursor proximity. Risk-behaviour will be manipulated across different conditions by manipulating a participant’s speed-accuracy trade-off and/or applying a penalty for crossing the target line. Cursor endpoint, variance, and total points are the primary measurements, allowing us to characterize the direction and magnitude of risk-behaviour. An optimal aimpoint model will be used to determine risk-sensitivity, with subjects' actual movement endpoints compared to their predicted movement endpoints found by testing their idiosyncratic motor noise. We hypothesize participants will more attentively consider their motor noise and reduce cursor excursion and endpoint variability when faster movements and/or penalty are introduced. Additionally, participants with a distorted perception of their motor accuracy would likely move closer/further than optimally predicted based on their own motor noise. Finally, risk-averse behaviour is more likely to be expressed following a penalized trial. If risk-behaviour is unaltered by the outcome of preceding trials, we assume that strategic changes are not made on trial-by-trial bases, but rather some average outcome of gains across trials.

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21. *3:33 – 3:39*

Clear Roles, Strong Teams: A Social Network Analysis of Role Clarity in Sport

Grace French (she/her)

Supervisor: Desmond McEwan

Roles are an important component of group functioning. They can be defined as a set of behavioural expectations for a specific position within a social structure. In a sporting context, understanding one's role in a team is critical toward individual performance and overall team success. Specifically, role clarity (opposite to role ambiguity) involves a clear understanding of one's role responsibilities within their team and how that role will contribute overall to team goals. A critical gap exists as role clarity has not yet been examined from an interpersonal perspective — that is, the shared understanding of an athlete's role as well as the role of their teammates.

Group and interpersonal constructs have been recently studied through an advanced methodology of network conceptualization. Although this approach has been leveraged to examine various constructs in team dynamics in sport, it has yet to be utilized to study role perceptions. Thus, the purpose of the proposed project is to examine interpersonal role clarity among athletes competing on interdependent teams with salient individual and team-level variables using social network analysis (SNA).

It is hypothesized that higher ratings of interpersonal role clarity will be correlated with higher ratings of (1) individual performance, (2) interpersonal role efficacy, (3) athletes' perceptions of team cohesion, and (4) athletes' perceptions of the team's teamwork. The proposed project offers a new layer within existing research of role perceptions in sport. Such work will enhance our understanding of this construct with empirical (e.g., network-based foundation) and real-world implications.

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22. *3:39 – 3:45*

Age-Related Differences in Vestibular Perception

Emmarie Racine Hallin (she/her), Josh Donald

Supervisor: Mark Carpenter

Vestibular perception enables the conscious awareness of head motion and is mediated by inner ear sensory organs and central processing pathways. Age-related deterioration of the vestibular system may impair motion perception. Prior work has largely focused on detection thresholds, while less is known about suprathreshold vestibular perception in aging. This study investigated whether suprathreshold perception of rotation about the yaw plane is altered in older adults.

Twenty-one healthy older adults (60-78 years) and twenty-two young adults (18-29 years) completed a rotation perception task. Participants tracked platform motion via a rotary dial during suprathreshold pseudorandom sinusoidal rotations (0.02-0.11 Hz; ±30°) with visual and auditory masking. Vestibular perception (VP) was quantified using coherence, gain, and phase between platform and dial responses. Peripheral vestibular function was assessed using the visual vestibulo-ocular reflex (VVOR), comparing eye and head velocity.

Reliability was evaluated using intraclass correlation coefficients (ICC). VP gain demonstrated good reliability (ICC > 0.70), whereas coherence showed poor reliability (ICC < 0.50). VP coherence, gain, and phase were significantly greater at high frequency compared to low frequency ($p < 0.05$). VP gain ($p = 0.016$) and phase also differed between age groups. In contrast, VVOR coherence did not vary by age or frequency. While reflexive VOR responses appear preserved with aging, suprathreshold vestibular perception — particularly gain and phase — shows age-related differences, suggesting deficits in central integration of vestibular information. VP gain emerged as the most reliable perceptual metric and may provide a sensitive measure of age-related vestibular decline.
implications.

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23. *3:45 – 3:51*

Optimizing Noisy Vestibular Stimulation for maximizing sway and increasing postural oscillations

Nada Shweiki, Liam Foulger

Supervisor: Jean-Sébastien Blouin

Human balance relies on the integration of visual, auditory, vestibular, and somatosensory inputs. To reveal the role of the vestibular system in balance control, electrical vestibular stimulation (EVS) has been widely used. Small currents delivered through surface electrodes over the mastoid processes induce well-defined balance responses in humans. When applied as noisy inputs, certain authors have proposed that low-amplitude electrical vestibular stimuli can enhance sensory processing and potentially improve postural control. Despite these promising findings, most researchers have applied fixed electrical stimulation parameters, overlooking inter-individual variability in vestibular sensitivity.

Here, we performed a pilot study to determine if a multidimensional optimization approach could identify electrical vestibular stimulation parameters that increase postural oscillations. Two healthy participants stood on force plates while receiving binaural bipolar electrical vestibular stimuli applied over their mastoid processes. The optimization approach iteratively adjusted stimulation parameters (0-2 mA; 0-640 Hz) based on real-time sway measurements. We quantified postural oscillations using centre-of-pressure (CoP) velocity.

After 6 generations of 18 parameter sets, our results showed that the approach successfully identified subject-specific stimulation parameters that increased postural oscillations. For the two participants, optimal parameters corresponded to (1.63 mA; 0.08–8.75 Hz), and (1.64 mA; 0.08–9.64 Hz), respectively. These preliminary data support the use of optimization parameters to identify stimulation parameters to modify postural oscillations. The next step will involve a complete study to identify electrical vestibular stimulation parameters that decrease postural oscillations, with the long-term goal to inform the development of personalized EVS-based wearable interventions to improve balance in everyday life.

Session 5 - 10-Minute Oral • 4:01 – 5:01 PM

23. *4:01 – 4:16*

What can we learn about the pelvic floor muscles on a (pseudo) roller coaster ride?

Xueqing Zhou (she/her/hers), Alison Williams, Jiho Song, Jeff Nickel, Tania Lam, Jean-Sébastien Blouin

Supervisors: Dr. Tania Lam, Dr. Jean-Sébastien Blouin

The pelvic floor muscles (PFM) form the base of the abdominopelvic cavity and are responsible for maintaining continence, especially during postural perturbation and dynamic movements. PFM contraction in these scenarios does not require volitional effort; rather, involuntary PFM activation occurs in response to increased intra-abdominal pressure and visceral loading due to gravity. Existing experimental paradigms to elicit involuntary PFM activity almost always involve co-activation of trunk and gluteal muscles, which subjects PFM electromyography (EMG) to crosstalk and limits our ability to determine the sources of neural drive acting on the PFM.

To address this gap, we assessed whether passive single-axis whole-body motion applied by a robotic arm (KUKA KR500-3) could evoke involuntary PFM response in the absence of confounding co-activation. We recorded surface EMG from the PFM, trunk, and gluteal muscles while the robotic arm moved healthy, continent participants through passive whole-body linear translation (anterior-posterior, lateral, vertical) or rotation (roll, pitch, yaw) trajectories. Each motion was repeated at 3 peak velocities with peak acceleration held constant.

In two pilot participants, motion-evoked PFM activation was present in response to all linear and >85% of rotary stimuli, while trunk and gluteal muscles generally remained quiescent. PFM activation primarily occurred during acceleration/deceleration. PFM EMG amplitude was positively correlated with peak velocity ($r=0.82$-0.88, $p<0.001$) during linear translations and pitch rotations. Our results support the use of passive whole-body motion stimuli to evoke involuntary PFM activity without confounding co-activation, providing a unique opportunity to isolate PFM activation from complex dynamic activities.

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24. *4:16 – 4:31*

Height-induced postural threat and its influence on upper limb fine motor control

Xiya (Christina) Cheng, Xiangwei Zhang

Supervisor: Mark Carpenter

Although standing balance seems fundamental for healthy young adults, it poses greater challenges for older adults, who experience increased fear of falling. Inducing postural threat through elevated height effectively elicits fear of falling under which individuals typically adopt a stiffening strategy characterized by frequent, low-amplitude adjustments of the center of pressure (COP) during quiet standing. While such change in postural control has been well documented in lower-limb control, less is known about how postural threat influences upper limb motor behavior. Upper limb goal-directed motor control requires the central nervous system to coordinate precise movement planning with anticipatory postural adjustments. However, anxiety can disrupt this process, particularly in complex tasks, by altering attentional focus and reducing processing efficiency.

The present study aimed to investigate the effects of height-induced postural threat and associated emotional responses on upper limb fine motor control, and explore the relationship between upper limb task performance and whole-body balance control. It was hypothesized that

increased fear would impair task performance, resulting in lower accuracy, even when the upper limb movement is examined independently from any underlying changes in postural control.

Twenty-three young adults completed three simple tracing tasks at both LOW (0.8 m) and HIGH (2 m) heights. Contrary to previous hypothesis, preliminary findings suggest a "stiffening" strategy in the wrist control, evidenced by reduced overshoot at HIGH compared to the LOW condition, indicating decreased movement variability and improved performance accuracy under threat. Furthermore, a clear association between upper limb performance and whole-body balance was observed, particularly in the circle tracing task, where COP fluctuations were temporally coupled with upper limb movements.

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25. *4:31 – 4:46*

Exploring Motivations for Attendance and Fanship in Canadian Women's University Sport

Ellie Saplywy (she/her/hers)

Supervisor: Dr. Andrea Bundon

Women's professional sport continues to grow at unprecedented rates in North America, with the addition of the Northern Super League, the Professional Women's Hockey League, and a Toronto-based Women's National Basketball Association team, all within the past three years. Recent research has found that the increase in professional women's leagues has demonstrated a substantial growth in fan investment and attendance for women's sports. However, comparable patterns are not reflected in women's university sport literature.

Currently no existing literature examines the motivations for, and potential benefits of attendance and fanship within university women's sport in a Canadian context. My proposed research aims to address this gap by expanding on my prior work related to exploring experiences of fanship within Canadian women's university sport. I plan to (1) identify factors that motivate or deter engagement with women's varsity sport through examining the development of fan identity in the Canadian women's university context; and (2) assess strategies to increase event attendance and support student wellbeing.

Adopting a critical feminist lens, my research design will include semi-structured interviews from both self-identified fans of university women's sport, as well as community members who do not currently engage with women's sports at UBC. Student well-being has emerged as a prominent institutional concern across Canada over the past two decades. Considering the prominent body of research linking team identification to psychological well-being, the outcomes of this research present an opportunity to explore how identification with women's sport might align with leverage support for student wellbeing within the context of the university.

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26. *4:46 – 5:01*

Computational models to explain the role of efferent copies in standing balance with delays

Amin MohammadiNasrabadi (He/Him/His), Pedro Amui, Izzy Zhang, Calvin Kuo, Jean-Sébastien Blouin

Supervisor: Jean-Sébastien Blouin

Sensorimotor delays pose a fundamental challenge to the control of standing balance. They introduce temporal mismatches between sensory feedback and motor actions that lead to postural instability and falls. Although humans can adapt to imposed delays of 400 ms, the mechanisms

underlying the adaptation process remain unknown. We previously showed that predictions of the sensory consequences of our actions could contribute to the adaptation processes underlying balancing with delays. Such deterministic controllers with forward models predicting our actions rely on explicit representations of the biomechanics of standing, which raises questions about biological plausibility.

To address this limitation and explore alternative ways to accommodate delays without the need for explicit predictions, we developed a computational Reinforcement Learning (RL) model that uses the Proximal Policy Optimization (PPO) approach to learn and control an inverted pendulum. We included the recent action history of motor commands (i.e. efferent copies) as an input state in addition to the angle and angular velocity of the human body. This augmentation transforms the delayed control problem into a higher-dimensional Markov decision process, enabling the policy to capture temporal dependencies implicitly.

Results demonstrate that the augmented RL controller substantially improves delay tolerance, maintaining stability at delays up to ~250 ms, compared to ~104 ms in the baseline RL model. The inclusion of efferent copies led to more stable and consistent control. Notably, the proposed approach achieved comparable delay compensation to deterministic models without requiring an explicit forward model. These findings suggest that incorporating efferent copies into RL provides a viable and biologically plausible mechanism for handling sensorimotor delays.

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